

Henry Harvin
YUVA INTERN PROGRAM

SQL DATA ANALYST - AGRIBUSINESS

Week 3 Project Report
Agribusiness Performance Analysis using SQL

Dataset:

Agribusiness Database - MySQL

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Scenario Development

Business Scenario

The agribusiness company manages several farms in different locations and grows different crops during different seasons and years. The business keeps records of cultivated area, production, crop yield, inputs, costs, and sales.

The management team wants to understand how well the farms and crops are performing. Production, yield, and costs can differ between crops, farms, seasons, and years. The company therefore needs to analyse its data to find these differences and identify areas that may need attention.

Business Problem

The management team wants to find out:

- Which crops give better production for the area used?
- How does yield differ between crops?
- How does production and yield change over the years?
- Do some seasons perform better than others?
- Which production activities have higher costs?
- How much does it cost to produce one unit?
- Which farms show differences in production performance?

Analysis Objective

The aim of this analysis is to use SQL to study the performance of the agribusiness operations. The analysis will focus on production efficiency, crop yield, yearly and seasonal trends, and operational costs. The results will be used to understand the main performance patterns and provide practical, data-based recommendations.

KPI Definition and Analysis Plan

To measure the performance of the agribusiness operations, the analysis will use a few key performance indicators (KPIs).

Production Efficiency

Production efficiency will be measured using yield per acre.

Formula:

Yield per acre = Production quantity ÷ Cultivated area

This helps show how much crop is produced from the land used.

Farm Performance

Farm performance will be studied using:

- Total production
- Average yield

This will help compare production levels and yield between farms.

Yearly Performance

Production and average yield will be compared across different years to see whether performance is changing over time.

Seasonal Performance

Production and average yield will be compared across seasons to identify differences in seasonal performance.

Operational Cost

Total production costs will be analysed to understand how much is being spent on different production activities.

Cost per Unit

Cost efficiency will be measured using:

Formula:

Cost per unit = Total production cost ÷ Production quantity

This shows how much it costs to produce one unit of crop.

Analysis Approach

The analysis will use SQL queries to calculate these KPIs from the production, crop, farm, season, and cost tables. The results will be shown using tables and charts, followed by simple explanations of the main findings.

Production Efficiency by Crop

The first analysis compares the production efficiency of different crops. Yield per acre is used to measure how much crop was produced from the cultivated area.

SQL Query

```
select
  c.crop_name,
  sum(p.production_quantity) as total_production,
  sum(p.cultivated_area) as total_area,
  round(sum(p.production_quantity) / sum(p.cultivated_area), 2) as yield_per_acre
from production p
join crop c
  on p.crop_id = c.crop_id
group by c.crop_name
order by yield_per_acre desc;
```

Results

Crop	Total Production	Total Area	Yield per Acre
Tomato	495.00	55.00	9.00
Sugarcane	640.00	80.00	8.00
Onion	240.00	30.00	8.00
Rice	905.00	175.00	5.17

Potato	275.00	55.00	5.00
Maize	520.00	110.00	4.73
Wheat	1175.00	270.00	4.35
Soybean	232.00	110.00	2.11
Cotton	230.00	115.00	2.00
Mustard	97.50	65.00	1.50

Explanation

The query groups the production records by crop and calculates the total production and total cultivated area for each crop. Yield per acre is then calculated by dividing total production by total cultivated area.

Tomato has the highest yield per acre at 9.00, followed by sugarcane and onion at 8.00 each. Mustard has the lowest yield per acre at 1.50.

Wheat has the highest total production at 1,175.00, but its yield per acre is 4.35. This shows that a crop can have high total production without having the highest production efficiency, because total production also depends on the amount of land used.

	crop_name	total_production	total_area	yield_per_acre
►	Tomato	495.00	55.00	9.00
	Sugarcane	640.00	80.00	8.00
	Onion	240.00	30.00	8.00
	Rice	905.00	175.00	5.17
	Potato	275.00	55.00	5.00
	Maize	520.00	110.00	4.73
	Wheat	1175.00	270.00	4.35
	Soybean	232.00	110.00	2.11
	Cotton	230.00	115.00	2.00
	Mustard	97.50	65.00	1.50

Farm Performance

The second analysis compares the farms based on their total production and average yield. This helps us see which farms produce more and which farms have better average yield.

SQL Query

```
select
    f.farm_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield
from production p
join farm f
    on p.farm_id = f.farm_id
group by f.farm_name
order by total_production desc;
```

Results

Farm	Total Production	Average Yield
Sunrise Agro Farm	1340.00	5.83
Ganga Plains Farm	745.00	4.83
Green Valley Farm	635.00	3.33
Haryana Fields Farm	562.50	5.33

Golden Harvest Farm	555.00	4.00
Nashik Harvest Farm	465.00	5.50
Agra Green Farm	375.00	5.00
Deccan Agro Farm	132.00	2.20

Explanation

The query groups the production records by farm and calculates the total production and average yield for each farm.

Sunrise Agro Farm has the highest total production at 1,340.00 and also has a high average yield of 5.83. Ganga Plains Farm has the second-highest total production at 745.00.

Nashik Harvest Farm has a total production of 465.00, but its average yield is 5.50, which is higher than several farms with greater total production.

Deccan Agro Farm has the lowest total production at 132.00 and the lowest average yield at 2.20. This shows that total production alone does not give the complete picture of farm performance.

	farm_name	total_production	average_yield
►	Sunrise Agro Farm	1340.00	5.83
	Ganga Plains Farm	745.00	4.83
	Green Valley Farm	635.00	3.33
	Haryana Fields Farm	562.50	5.33
	Golden Harvest Farm	555.00	4.00
	Nashik Harvest Farm	465.00	5.50
	Agra Green Farm	375.00	5.00
	Deccan Agro Farm	132.00	2.20

Yearly Performance

The third analysis compares production and average yield across the two years in the dataset. This helps us see whether overall performance changed over time.

SQL Query

```
select
  production_year,
  sum(production_quantity) as total_production,
  round(avg(yield_quantity), 2) as average_yield
from production
group by production_year
order by production_year;
```

Results

Year	Total Production	Average Yield
2024	1942.50	3.39
2025	2867.00	5.70

Explanation

The results show that total production increased from 1,942.50 in 2024 to 2,867.00 in 2025. Average yield also increased from 3.39 to 5.70.

This shows that the farms recorded higher overall production and better average yield in 2025 compared with 2024. The increase in average yield suggests that the higher production was not only due to the amount of production recorded, but was also accompanied by better yield across the production records.

	production_year	total_production	average_yield
▶	2024	1942.50	3.39
	2025	2867.00	5.70

Seasonal Performance

The fourth analysis compares production and average yield across different agricultural seasons.

SQL Query

```
select
    s.season_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield
from production p
join season s
    on p.season_id = s.season_id
group by s.season_name
order by total_production desc;
```

Results

Season	Total Production	Average Yield
Kharif	2527.00	4.12
Rabi	1547.50	3.71
Summer	735.00	8.67

Explanation

Kharif has the highest total production at 2,527.00, followed by Rabi at 1,547.50. However, Summer has the highest average yield at 8.67, even though its total production is lower at 735.00.

This shows that total production and average yield can give different views of performance. Kharif produces more overall, while Summer has a higher average yield across its production records.

	season_name	total_production	average_yield
►	Kharif	2527.00	4.12
	Rabi	1547.50	3.71
	Summer	735.00	8.67

Operational Cost Analysis

The fifth analysis looks at the different types of operational costs recorded for the production activities. This helps identify which areas are taking up more of the total cost.

SQL Query

```
select
    cost_type,
    sum(amount) as total_cost
from cost
group by cost_type
order by total_cost desc;
```

Results

Cost Type	Total Cost
Labour	96,500.00
Machinery	30,000.00
Irrigation	29,300.00
Transportation	12,600.00

Explanation

Labour has the highest total cost at 96,500.00, which is much higher than the other cost types. Machinery costs 30,000.00, while irrigation costs 29,300.00. Transportation has the lowest cost at 12,600.00. The results show that labour is the main operational cost in the available data. This makes labour an important area to consider when looking for ways to control or improve production costs.

	cost_type	total_cost
▶	Labour	96500.00
	Machinery	30000.00
	Irrigation	29300.00
	Transportation	12600.00

Cost per Unit of Production

The sixth analysis looks at the cost of producing each production record. Cost per unit is calculated by dividing the total production cost by the production quantity.

SQL Query

```
select
    p.production_id,
    c.crop_name,
    p.production_quantity,
    sum(co.amount) as total_cost,
    round(sum(co.amount) / p.production_quantity, 2) as cost_per_unit
from production p
join crop c
    on p.crop_id = c.crop_id
join cost co
    on p.production_id = co.production_id
group by
    p.production_id,
    c.crop_name,
    p.production_quantity
order by cost_per_unit desc;
```

Results

Production ID	Crop	Production Quantity	Total Cost	Cost per Unit
2	Mustard	45.00	4,500.00	100.00

11	Mustard	52.50	4,800.00	91.43
5	Cotton	80.00	7,000.00	87.50
18	Cotton	150.00	12,000.00	80.00
14	Soybean	100.00	6,500.00	65.00
20	Soybean	132.00	7,500.00	56.82
1	Wheat	320.00	14,000.00	43.75
6	Potato	125.00	5,000.00	40.00
7	Wheat	450.00	16,500.00	36.67
13	Wheat	405.00	14,500.00	35.80
17	Potato	150.00	5,100.00	34.00
3	Maize	270.00	8,500.00	31.48
4	Rice	350.00	10,800.00	30.86
12	Tomato	180.00	5,200.00	28.89

8	Maize	250.00	7,000.00	28.00
16	Rice	225.00	6,200.00	27.56
10	Rice	330.00	9,000.00	27.27
15	Onion	240.00	5,300.00	22.08
19	Tomato	315.00	6,500.00	20.63
9	Sugarcane	640.00	12,500.00	19.53

Explanation

The results show that cost per unit varies considerably between production records. Production ID 2, which is mustard, has the highest cost per unit at 100.00. Production ID 9, which is sugarcane, has the lowest cost per unit at 19.53.

The results also show differences within the same crop. For example, the two mustard records have cost per unit values of 100.00 and 91.43. This means that production cost can vary even when the crop is the same.

This measure is useful because looking only at total cost does not show how efficiently the money is being used. Cost per unit gives a clearer view of the cost required to produce each unit of output.

	production_id	crop_name	production_quantity	total_cost	cost_per_unit
▶	2	Mustard	45.00	4500.00	100.00
	11	Mustard	52.50	4800.00	91.43
	5	Cotton	80.00	7000.00	87.50
	14	Soybean	100.00	6500.00	65.00
	6	Potato	125.00	5000.00	40.00
	20	Soybean	132.00	7500.00	56.82
	18	Cotton	150.00	12000.00	80.00
	17	Potato	150.00	5100.00	34.00
	12	Tomato	180.00	5200.00	28.89
	16	Rice	225.00	6200.00	27.56
	15	Onion	240.00	5300.00	22.08
	8	Maize	250.00	7000.00	28.00
	3	Maize	270.00	8500.00	31.48
	19	Tomato	315.00	6500.00	20.63
	1	Wheat	320.00	14000.00	43.75
	10	Rice	330.00	9000.00	27.27
	4	Rice	350.00	10800.00	30.86
	13	Wheat	405.00	14500.00	35.80
	7	Wheat	450.00	16500.00	36.67

Overall Crop Performance

The final analysis combines production, average yield, and cost information for each crop. This gives a broader view of crop performance and helps compare output with the cost involved.

SQL Query

```
select
    c.crop_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield,
    sum(co.amount) as total_cost,
    round(sum(co.amount) / sum(p.production_quantity), 2) as cost_per_unit
from production p
join crop c
    on p.crop_id = c.crop_id
join cost co
    on p.production_id = co.production_id
group by c.crop_name
order by cost_per_unit;
```

Results

Crop	Total Production	Average Yield	Total Cost	Cost per Unit
Sugarcane	1920.00	8.00	12,500.00	6.51
Onion	480.00	8.00	5,300.00	11.04

Tomato	990.00	9.00	11,700.00	11.82
Rice	2160.00	5.14	26,000.00	12.04
Wheat	3525.00	4.33	45,000.00	12.77
Maize	1040.00	4.75	15,500.00	14.90
Potato	550.00	5.00	10,100.00	18.36
Soybean	464.00	2.10	14,000.00	30.17
Cotton	610.00	2.00	19,000.00	31.15
Mustard	195.00	1.50	9,300.00	47.69

Explanation

Sugarcane has the lowest cost per unit at 6.51 and an average yield of 8.00. Tomato has the highest average yield at 9.00, while its cost per unit is 11.82.

Wheat has the highest total production at 3,525.00, with a cost per unit of 12.77. Rice also has high total production at 2,160.00 and a cost per unit of 12.04.

Mustard has the highest cost per unit at 47.69 and the lowest average yield at 1.50. Cotton and soybean also have relatively high costs per unit compared with the other crops.

The results show that looking at production alone is not enough to understand crop performance. Comparing production, yield, and cost together gives a clearer picture of how efficiently each crop is being produced.

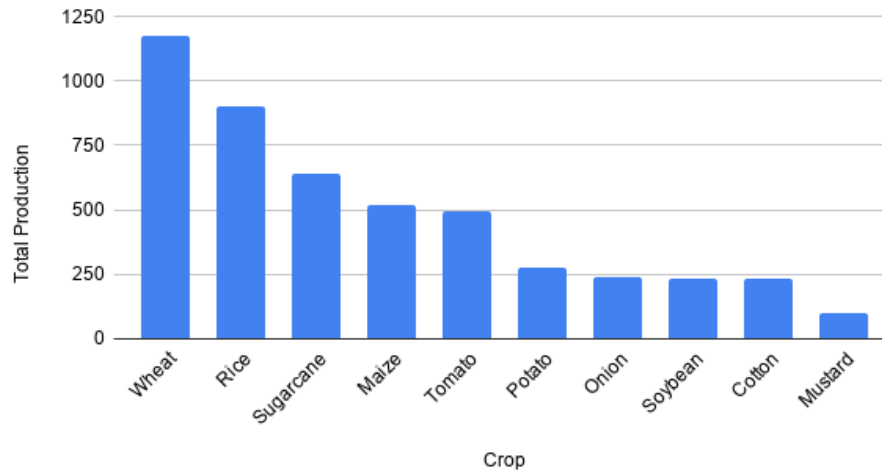
	crop_name	total_production	average_yield	total_cost	cost_per_unit
►	Sugarcane	1920.00	8.00	12500.00	6.51
	Onion	480.00	8.00	5300.00	11.04
	Tomato	990.00	9.00	11700.00	11.82
	Rice	2160.00	5.14	26000.00	12.04
	Wheat	3525.00	4.33	45000.00	12.77
	Maize	1040.00	4.75	15500.00	14.90
	Potato	550.00	5.00	10100.00	18.36
	Soybean	464.00	2.10	14000.00	30.17
	Cotton	610.00	2.00	19000.00	31.15
	Mustard	195.00	1.50	9300.00	47.69

Visual Analysis

The charts below present the main production, yield, yearly performance and cost results obtained from the SQL analysis. They make it easier to compare the performance of different crops and identify important differences in production and operational costs.

Total Production by Crop

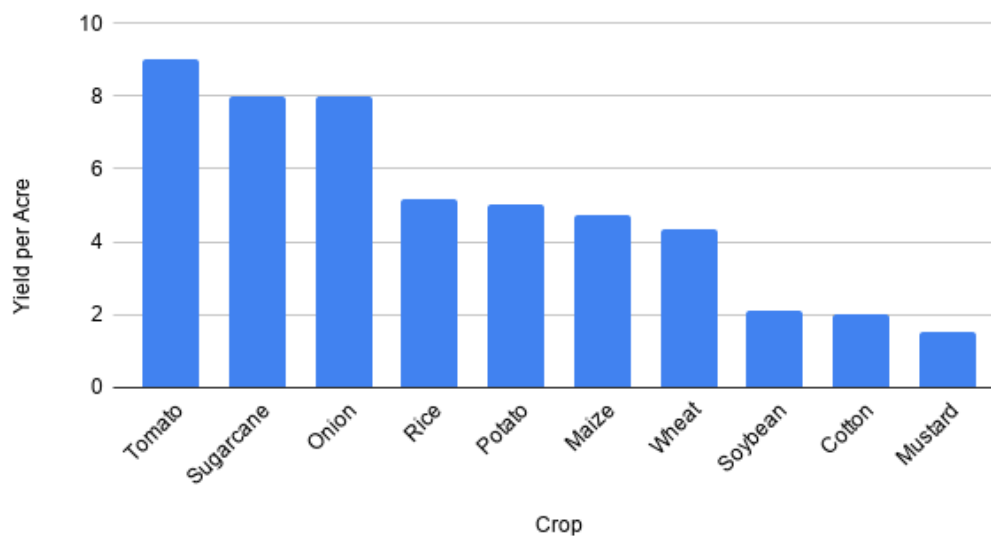
Total Production by Crop



The chart shows that wheat has the highest total production at 1,175 units, followed by rice at 905 units and sugarcane at 640 units. Mustard has the lowest total production at 97.5 units. This shows that total production varies considerably between crops.

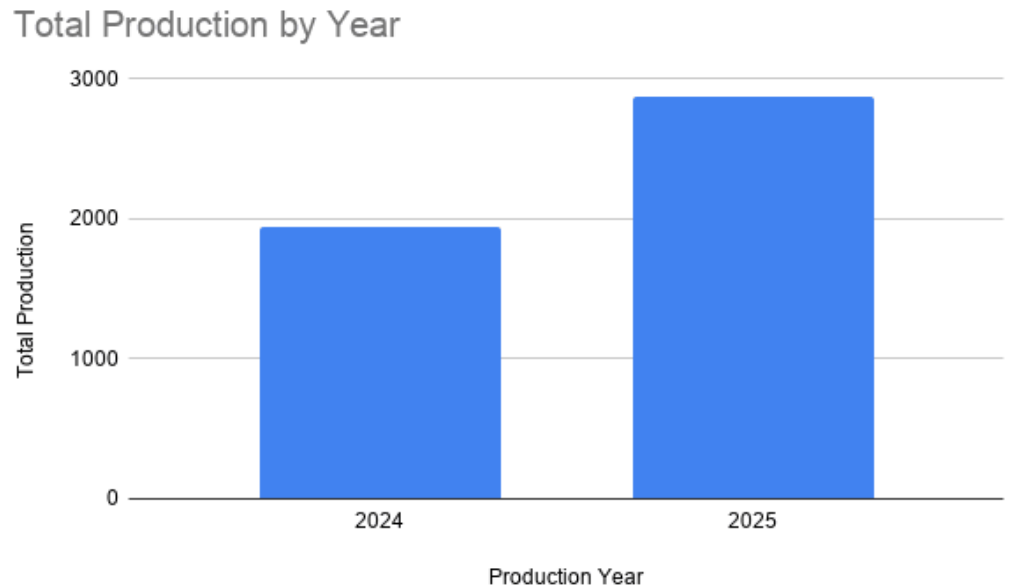
Yield per Acre by Crop

Yield per Acre by Crop



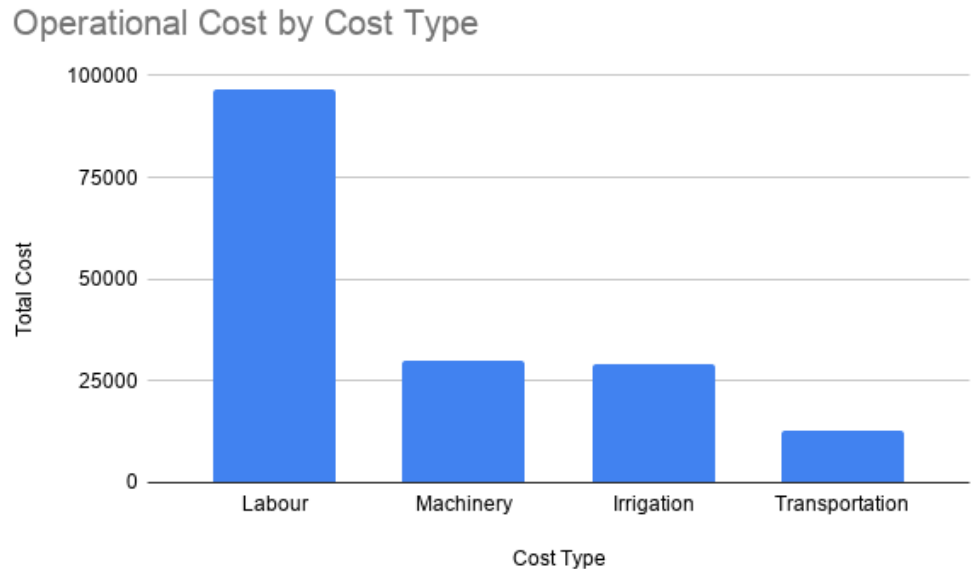
The chart shows differences in production efficiency between crops. Tomato has the highest yield per acre at 9.00, followed by sugarcane and onion at 8.00 each. Mustard has the lowest yield per acre at 1.50. This shows that a crop with high total production does not always have the highest yield efficiency.

Total Production by Year



Total production increased from 1,942.50 units in 2024 to 2,867.00 units in 2025. This shows an increase in overall production between the two years in the dataset.

Operational Cost by Cost Type



Labour is the largest operational cost at 96,500, followed by machinery at 30,000 and irrigation at 29,300. Transportation has the lowest cost at 12,600. The difference shows that labour is a major part of the operational cost structure.

Insights and Business Interpretation

The analysis gave some useful findings about the farms, crops, production and costs.

Different crops have different yields

Tomato has the highest yield per acre at 9.00. Sugarcane and onion have a yield of 8.00. Mustard has the lowest yield at 1.50. Wheat has the highest total production at 1,175 units, but its yield per acre is 4.35. This shows that the crop with the highest production is not always the most efficient.

Farm performance is different

Sunrise Agro Farm has the highest production at 1,340 units and an average yield of 5.83. Nashik Harvest Farm has lower production at 465 units but still has a good average yield of 5.50. Deccan Agro Farm has the lowest production at 132 units and the lowest average yield at 2.20.

Production increased in 2025

Total production increased from 1,942.50 units in 2024 to 2,867 units in 2025. Average yield also increased from 3.39 to 5.70. So, the farms had better production and yield in 2025.

Seasons show different results

Kharif has the highest total production at 2,527 units. Summer has the highest average yield at 8.67. This means that Kharif produced more overall, but Summer had better yield per acre.

Labour is the biggest cost

Labour is the highest cost at 96,500. Machinery costs 30,000, irrigation costs 29,300, and transportation costs 12,600. This shows that labour takes up a large part of the total operational cost.

Cost per unit is different for each crop

Sugarcane has the lowest overall cost per unit at 6.51, while mustard has the highest at 47.69. Sugarcane also has a good average yield of 8.00, while mustard has the lowest average yield of 1.50. This shows that both yield and cost are important when looking at crop performance.

Overall, the results show that production, yield, and cost can be quite different across crops, farms, and seasons. Looking at all these measures together gives a better picture of the overall performance.

Actionable Recommendations

Based on the analysis, the business can:

- Focus more on crops with better yield and lower cost per unit.
- Check the high labour cost and look for ways to reduce unnecessary labour expenses.
- Review crops like mustard that have low yield and high cost per unit.
- Study the better-performing farms and seasons to understand what is working well.
- Keep tracking production, yield, and costs every year to compare performance.

Conclusion

This analysis helped understand the performance of the agribusiness data using SQL. The analysis looked at crop production, yield, farm performance, seasonal trends, and operational costs.

The results showed that performance is different across crops, farms, and seasons. Production and average yield also increased in 2025, while labour was the biggest operational cost.

Overall, using SQL made it easier to find these patterns and compare different parts of the business. The results can help the business understand its performance and identify areas that may need improvement.